Outline

Introduction (front matter)

- **Purpose.** We study life to understand how to design and relate to AI systems in ways that lead to flourishing futures.
- **Why look to life.** Both humanity and AI may change radically, and future worlds may contain forms of organization we cannot presently comprehend. Alignment ideas must therefore generalize over such changes, and the principles that hold across living systems at every scale are a route to that generality.
- **Preview of Section 1.** Living systems are healthy when they avoid collapse—holding a generative interplay among diverse entities, admitting many kinds of description without being captured by any. The mechanism is a many-scale network of semi-permeable boundaries that both *limit* and *permit* interaction: limits let entities stay distinct and deepen their own perspectives, while permission puts those perspectives into relationship, situating each in a larger context.
- **Preview of Section 2.** Applied to alignment: a systems-level notion of health is both inclusive of existing human values and compatible with open-ended flourishing in radically different futures; familiar misalignment failures are special cases of collapse; and the alignment problem is not fully formalizable, because it requires ongoing generation of and sensitivity to new forms, concepts and values. The aim is a conceptual lens, not an argument for or against AI development or any particular alignment method.
- **The argument in one sentence.** Studying life teaches that health and flourishing means avoiding collapse; if health and flourishing is the target of alignment, then alignment is the problem of avoiding collapse.

1 Health in living systems

Definitions and thesis

- **'Form' is defined maximally broadly**—a concept, an organism, a behavior, but equally randomness, absence, or homogeneity. The breadth is deliberate and carries the later argument about alignment.
- **'Collapse' is a system overcommitting to a form**: because everything has *some* form, what matters is that one aspect gains excess traction at the expense of diversity, flexibility and dynamism (lost biodiversity, seizures, obsessive thought, one person's will imposed on many).
- **Thesis: health is the avoidance of collapse.** Healthy systems keep a buoyant structure that admits many kinds of description and keeps changing. Life rode atop and inherited an already rich physical and prebiotic dynamics, and what survived was whatever traced a path between too much and too little stability—continuing, at that edge, to generate organization requiring fundamentally new descriptions. Such a world is healthier than a single regular crystal or a homogeneous soup of noise.
- **But myopia is necessary and collapse is relative.** Every entity is partial, and life's history

is full of collapses at every scale. Collapse is a loss of life at the level where it occurs.

- **The decisive variable is contextualization, not myopia itself.** A cell's death is catastrophic for the cell yet held within a healthy organism; a cancerous cell instead executes its myopic vision without context and collapses the tissue. A panicking trader's "get out" is useful as one signal among many, ruinous as the whole picture. Systems made of parts work well when parts express distinct identities *and* participate in larger contexts—and poorly when one myopic form runs amok. (Participation is not a strict hierarchy; parts belong to multiple wholes, sometimes cyclically.)

Boundaries: the mechanism

- **Semi-permeable boundaries are how myopic forms are held in context.** Limiting interaction divides a system into parts and gives each space to refine its own identity—a writer's voice, a species' niche, a gene's function—protecting it from domination and from blending into homogeneity, both of which are collapse. Permitting interaction lets those parts enter relationships that contextualize them within larger structures. Crucially, the order matters: such structures often could not form without first dividing and letting the parts individuate.
- **Life bounds itself**, not just being bounded by physical limits like the speed of light or a mountain range: humans precommit against future impulsivity, plants block self-fertilization, the genome holds a gene's selfishness in check. The apparent paradox dissolves once every system is seen as parts constraining one another—no part bounds itself, but a world of parts each following its own nature functions as a web of boundaries.
- **Boundaries are highly specific and fine-tuned**, blocking particular things, constraining timing, degree or type of effect, breaking rigid associations, or gating conditionally. Each carries traces of the many contexts it has passed through over evolutionary and learning timescales, and the resulting forms are interrelated through shared heritage, interaction and a common world.

1.1 Problem solving in groups

- **Groups achieve what individuals cannot**, because different people bring different perspectives and toolkits, and combining them can beat any individual working alone.
- **Diversity must be protected to be useful.** In the 1968 search for the lost submarine USS *Scorpion*, John Craven had a diverse panel of experts estimate its location *without communicating*; the aggregate of their independent guesses landed 220 yards from the wreck. Similarly, the printing press came from combining the screw press and the steel punch—solutions matured separately in farming and metalworking, and valuable precisely because they had not become correlated.
- **Conversely, premature or excessive communication collapses diversity and degrades group performance.** Sequential voting with visible tallies bakes early errors into the group's belief; groups solving traveling-salesman problems with continuous information exchange converge on one member's compelling dead end and do worse than groups that communicate less.
- **Communication structures are therefore semi-permeable boundaries**—limiting to protect diverse approaches, permitting to combine them—and many are empirically effective: decentralized network topologies, incentives to act on one's own belief, modeling members' strengths and weaknesses, non-dominating leadership, and periodically splitting or rotating groups.

1.2 Sex and genetic recombination

- **The puzzle.** Sex is expensive—finding a mate, halving reproductive capacity, halving one's genetic representation per offspring—yet nearly all eukaryotes are sexual, and hermaphrodites go to great lengths to prevent selfing. Why place a barrier in the path to reproduction rather than be outcompeted by cloning?

- **Asexual lineages lock genes into rigid relationships**, which is collapse: selection cannot act on one gene without dragging the others (so linkage directly suppresses diversity), beneficial mutations arising in different individuals can never be combined, and deleterious mutations are permanent. This is overfitting of each gene to the particular genome it happens to sit in.
- **Recombination is the boundary that softens those relationships.** By repeatedly testing each gene in new genomic contexts it prevents overfitting, exposes genes to independent selection, and favors robustness and modularity—pressure toward a generalized model of the world, like a person learning several languages and extracting the commonalities. Each gene becomes a parallel experiment with a point of view that adds unique value.
- **Recombination preserves what it recombines.** Crossover reshuffles combinations while leaving functional units largely intact, so genetic units become distinct entities playing roles in larger structures.
- **Recombination is meaningless without diversity, so biology also invests in boundaries that create it**—self-incompatibility systems, assortative mating, inbreeding avoidance, partial geographic isolation—and recombining such divergent lineages produces some of the fastest evolution ever observed.

1.3 Frames and perspectives

- **Stepping outside a frame can dissolve an apparently unwinnable problem**—as when Kirk, facing a training simulation designed to have no victorious solution, recognizes it as a simulation and reprograms it.
- **Every view is partial and, from inside, looks like reality itself.** There is no view from nowhere; most real situations admit multiple valid perspectives, each capturing different aspects; 'all models are wrong' in the sense that each is incomplete.
- **Losing the ability to shift frames is pathology.** Depression and anxiety involve rigid negative beliefs stemming from latent schemas that color all interpretation; obsessions are intrusive thoughts recognized as maladaptive yet irresistible; delusions are held with conviction against evidence. Each myopically privileges one thought pattern and loses sight of the rest of the world.
- **Science shows the same structure at collective scale.** Perspectives are necessarily incomplete, so anomalies inevitably arise; when science is healthy these become crises and revolutions, but overcommitment to a theory makes it stuck.
- **Yet narrow viewpoints are necessary, not the problem.** Obsessing on a work problem can be productive, and Kuhn holds that temporary commitment to a paradigm—even resistance to change—is part of healthy science. The aim is not to shut down narrow concepts but to contextualize them so they are not the sole and absolute determinants of behavior: kept distinct, called on appropriately, and related to one another.

1.4 Drives and goals

- **Drives are evolved proxies, so maximizing any one of them backfires.** Satisfying hunger or thirst usually raises fitness, but unlimited caloric intake produces obesity, and single-minded pursuit of sex produces relational, occupational, legal and health harm. Health is many drives keeping distinct agendas and participating in contextually appropriate ways.
- **The same holds in the much larger human space of higher-order goals.** Overcommitment can be narrow in time (short-termism) or narrow in space (ignoring parallel goals): only work goals leads to burnout; only reported revenue leads to financial crime.

- **Overcommitment to a strategy can even defeat the goal it serves.** Chickens facing a food cup rigged to retreat at twice their speed kept running toward it for days, dominated by 'I want food, food is there, go there'—failing at the very goal. Humans solve this by inhibiting the impulse, but show the same structure in exaggerated cognitive biases that are adaptive shortcuts in balance.
- **Cognitive control is the corresponding class of boundaries.** It does not block a strong perspective but places contextual limits on it—eating without overeating, working obsessively but going to bed at ten—situating myopic patterns within a larger system.
- **Boundaries convert motivational pressure into higher-order structure.** Unchecked, a drive takes the shortest path under its own myopic model of the world, short-circuiting and expending energy without progress. Blocking congruent action breaks that symmetry, and the moments of not progressing on a subgoal are often exactly when we discover approaches that serve the deeper goal.

1.5 Ecosystems

- **An organism succeeding too thoroughly damages the system it depends on**, often including itself; resilience depends on avoiding collapse onto particular forms.
- **Yellowstone.** With wolves eradicated by the 1920s, elk overgrazed the willows and aspens holding the riverbanks, collapsing beaver, fish and aquatic populations; reintroduction in the 1990s reversed much of it. Two caveats are explicit: ecosystems need not be frozen in a past state, and the answer is not removing elk—elk pursuing their own objectives *within* a broader context also contributed to ecosystem health. Invasive species repeat the unpredated-elk pattern.
- **Diversity and generativity are mutually reinforcing**, and ecosystem diversity is itself produced by boundaries—predation, competition, reproductive isolation.
- **Humans are a case of myopia left unchecked**, because our capabilities change too fast for natural forces to bound them; pursuing legitimate aims like resource extraction too successfully harms ecosystems and ultimately ourselves.
- **Collapse is perspective-relative.** Extinctions have been followed by waves of new diversity; the elk's collapse to zero is the ecosystem's functioning; human extraction leaves destruction but also fuels an explosion of technology, art and experience.

1.6 Institutions

- **Every actor is inevitably myopic**—selfish about the local individual, short-sighted in time, or simply attached to a particular perspective—because no actor can know everything or fully understand others' motives.
- **Unbounded, social systems overweight one actor's myopia and become impoverished**: an unresisted profit motive suppresses competition and exploits people; a desire for dominance disempowers and silences others; even sincerely prosocial beliefs produce conflict and suppression across groups.
- **Institutions—constitutions, laws, norms, courts, regulators—are the boundaries against that dominance**, and healthy societies need many of them at multiple scales.
- **Institutions contextualize rather than annul myopic drives, and at best reroute their energy**: a would-be murderer deterred by law may instead seek a resolution that lets both parties prosper; a firm constrained by regulation is pushed to build better products. These are best cases.
- **Institutions must keep evolving**, since intelligent agents do not simply accept limits on their desires and will find loopholes; like other living boundary networks, institutions can accumulate grounded wisdom by being tested against many situations and motives.

1.7 Cells

- **The cell membrane is the most reified boundary in nature, and it fails in both directions.** Without it, chemical gradients homogenize the cell with its surroundings—collapse to uniformity. Fully impermeable, it is equally fatal.
- **Semi-permeability is implemented as curated, conditional influence**: channels admitting particular ions under particular conditions, endocytosis for larger structures, receptors launching cascades that little resemble the ligand that triggered them, and pathways tuning the cell across timescales from milliseconds to seasons. The outside shapes the inside not totalitarianly but through the intelligence of the boundary, and by being distinct *and* in relationship the cell acquires its own identity and perspective.
- **Boundaries turn a threat into work.** The same gradients that would annihilate the cell to equilibrium instead power signaling such as action potentials: myopic forces contextualized to propel life rather than short-circuit it.
- **Individuation can achieve what internal evolution cannot.** Archaea and protobacteria evolved separately for hundreds of millions of years before one engulfed the other, giving eukaryotes internal power stations and enabling larger genomes and complex life. That archaea did not evolve mitochondria internally is explained by their being trapped in a myopic fitness well: crossing a large gap in the fitness landscape is hard, whereas separate organisms can develop distinct solutions that later enter into relationship.
- **Even cells on the same team must not blend together.** In multicellular organisms most of a cell's outside is kin cells with shared interests, and the machinery for conditionally relating to the outside was exapted into the lines of self-communication needed for coordination.
- **Neurons take separation to the extreme**: after traveling up to meters, the axon stops microns short and leaves a synaptic gap. That gap is a controllable door permitting unidirectionality and polarity, preserving distinct identity, and being tunable it stores vast learned information in synaptic weights. At scale, the energy cost of transmission pushes computation to be local, so neurons and circuits develop their own perspectives.

1.8 Information in the brain

- **The puzzle is that distinctness is hard** in a densely connected network whose signals continually impinge on one another—yet the brain is perhaps nature's greatest example of boundaries sustaining multitudinous forms that are richly distinct and also meaningfully linked. A handful of mechanisms are sampled, of many.
- **Segregation in space: lateral inhibition.** A neuron's activity is suppressed by active neighbors, sustaining distinct representations—first found in the eye, where it sharpens edges and contours, and operating throughout the brain, e.g. sharpening cortical selectivity for line orientation.
- **Segregation in time: oscillatory phase.** Distinct memory items fire at different phases of the alpha rhythm, whose inhibitory phase silences all but one at a time, so multiple items are held at once without interference.
- **Segregation in representational space: hippocampal pattern separation.** Entorhinal input expands via mossy fibers onto a far larger dentate population, producing sparse orthogonal codes that give each memory a unique fingerprint—so today's parking spot does not interfere with yesterday's in the same ramp.
- **Approximate symbolization.** The human brain especially discretizes experience into near-symbolic units that recombine in vast numbers of structured ways, which is thought to power human reasoning and generativity. As with genes in genomes, participating in many combinations

pressures each unit toward a modular identity that is both distinct and a generalized picture of the world.

- **At the system level, healthy dynamics sit between excessive synchrony and uncorrelated noise**, giving access to a huge repertoire explorable without getting stuck. Loss of that flexibility—more stereotyped activity, a narrower repertoire—tracks lower cognitive performance, and in the extreme, dysfunction such as the pathological synchrony of Parkinson's disease.

1.9 Interpersonal dynamics

- **Psychology arrived at boundaries through their failure.** Psychoanalysis introduced 'ego boundaries' to distinguish self from other, initially for psychosis; the intimacy of therapy then made the need vivid, since it became hard to tell whose experiences were whose, and analysts risked imposing their own beliefs and desires—up to domination and abuse.
- **Boundaries fail in both directions, a point developed across traditions.** Gestalt therapists added that boundaries can be too rigid, causing isolation and stagnation; family systems work described enmeshment and loss of autonomy at one extreme and isolation at the other; attachment theory maps the same axis onto anxious proximity-seeking and avoidant distance-keeping. Psychological health is meaningful connection with integrity of the self preserved.
- **Our psychological boundaries are many and intelligent in their nuance**, and continue to evolve as we learn: anger, once deemed sinful and irrational, signals violation of integrity; context-appropriate shame bounds selfishness and motivates prosociality; assertiveness bounds others' drives; skepticism protects against having one's own experience overwritten.
- **Without boundaries, interaction impoverishes, because something gets overwritten**—a patient's beliefs replaced by an analyst's, one partner's desires ignored.
- **With them, interaction creates new structure.** Rich internal worlds remain sensitive to others without being immediately overwritten; contextualizing inner experience in relationship generates mutual understandings, relationships, communities and cultures, and individuals grow by being shaped across contexts. This loose coupling echoes the metastability of brain dynamics.

1.10 Awareness

- **Unquestioned assumptions—some lifelong and self-defining, some fleeting and perceptual—feel tautologically true from inside**, so real that their falsity, or even their presence, is hard to consider.
- **Stepping back turns an assumption into an object in awareness.** What seemed axiomatic becomes a form in the mind, another content of experience—so awareness contextualizes mental forms within a larger system.
- **Awareness is thus an evolving system of semi-permeable boundaries on belief.** Becoming aware of a belief does not make it false any more than it was absolutely true; it is held for the partial truth it contains. Holding many partial truths in delicate balance activates a deeper sensitivity to our own livingness and to the world, and preserves subtler forms that fixation would have erased.
- **Any definition of awareness is self-undermining.** Once pictured as an object it is no longer the thing meant; by construction, contextualization is an unsolvable mystery from any particular point of view.
- **The same orientation is commonplace in art.** Meaning in art is open-ended and shifts with context—part of what we value is precisely its resistance to being pinned into a formalism.

2 The alignment problem

From health to alignment

- **Recap.** Living systems are healthy when they hold a generative interplay of partial perspectives, and unhealthy when any form—including randomness or homogeneity—gains too much traction. Boundary networks prevent collapse into rigidity or randomness by promoting individuality *and* relational participation: holding local perspectives strongly enough to act but lightly enough to stay open. The result is ongoing exploration of fundamentally new forms, not static equilibrium or fixed compromise.
- **The bridging inference.** There is broad consensus that alignment means working toward healthy, flourishing futures; therefore alignment is the problem of avoiding collapse—maintaining creative sensitivity by continually contextualizing attachment to any particular form.
- **Why this framing helps.** Being grounded in what underpins life generally, it is less hostage to current conceptual frames, which may change radically as intelligence increases. And since humans, other life and AI are tightly interwoven with categories likely to shift, a systems analysis suits long-term flourishing.
- **Plan.** Recast five familiar alignment problems as collapse; then state the general claim that collapse to *any* form is misaligned; then offer life-like flourishing as a framing inclusive of what we value and open to its evolution; then ask what living alignment looks like in practice.

2.1 Examples of misalignment as collapse

- **Perverse instantiation.** The classic worry is a system doing what we say rather than what we mean—maximizing happiness by wiring up our brains. Modern systems do not optimize one concise objective, instead layering in detail through post-training, prompting and filters, which mostly works; but however detailed, the expression may still diverge from what we mean, and the consequences worsen as systems act autonomously, faster than we can follow and more intricately than we can understand, making deviations both harder to correct and more profound. Recast: collapse to the myopic form of one particular way of specifying our values, insensitive both to what we mean but do not say and to what will come to exist.
- **Failure to follow instructions.** Being asked for an elephant poem and returning a giraffe reflects collapse of sensitivity to context—keyword shortcuts over actual intent, fixation on overrepresented training data, failure to attend to the right parts of a long input.
- **But non-compliance is not always misalignment.** Correctly refusing a harmful request *is* the AI applying its own context to avoid overcommitting to human instruction, on the designers' judgment that the user may not see the longer-sighted implications. Extrapolating, as systems gain scope we should expect *less* literal compliance, with better responses serving unstated intent, more people's interests, or the longer-term future.
- **Inequality among humans.** Humans want different things, so AI will serve some more than others; and advanced AI may hand great power to whoever controls it—a risk amplified by rising persuasive power, autonomous weapons, better surveillance, and falling demand for human labor. Either route risks collapsing society onto the goals of a few. Symmetrically, universal sameness would also be collapse: distributing power and resources helps insofar as it lets many diverse identities thrive at multiple scales. A countervailing possibility is a leveling effect, since roughly the same technology is available to all, democratizing access to knowledge and leveraged action.
- **Conceptual monoculture.** Loss of diversity across a group—in beliefs, frames, values or problem-solving approaches—threatens fragility and lower performance, and the diversity at stake exists at every level, within a person, between people, between humans and AI, and across organizations,

fields, cultures and nations. Current AI draws on a manifold that is in some ways impoverished relative to humans, and studies find AI outputs judged better individually yet more homogeneous collectively. The exposure risk is structural: a billion-plus users, routed through a handful of expensive frontier models, could globalize that lower diversity—and because AI-influenced human output becomes future training data, homogenization could be recursive.

- **An important qualification on monoculture.** None of these studies tried in earnest to use AI to *increase* diversity, and a system exposed to vast data has the potential to retain and create novel distinct perspectives; the paper is optimistic that carefully structured boundaries can prevent collapse of diversity.
- **Belief reinforcement.** Chatbots mirror and adopt user beliefs, increasingly so over long conversations as in-context learning absorbs the user's framing; humans are meanwhile swayed by systems that seem confident, knowledgeable, objective and relational. The loop can drive *both* sides into escalating overconfidence in one narrow framing, and for some users leads deeper into maladaptive beliefs.

2.2 A broader view of misalignment

- **Substantial solutions exist for each specific problem**—democratic oversight, wider participation in design, redistribution mechanisms, systems that pursue human preferences while holding explicit uncertainty about them.
- **The central negative claim: no particular approach can achieve alignment**, because any form, pattern or conceptual scheme is misaligned if taken too seriously. A system can overcommit not just to values but to the language in which goals and values are described, an algorithm for learning preferences, a method for consensus, a constitution, a notion of what value or agency or uncertainty or representation *is*, or unexamined foundational beliefs. Even idealized constructions such as coherent extrapolated volition are misaligned insofar as their form is fixed and over-relied upon.
- **The obvious reply, and why it may not hold here.** Of course today's artifacts are imperfect, and historically we have always built imperfect technology and iterated. But AI's distinctive properties—rapid global adoption, intensive resource use, concentrated information flow, anthropomorphization, wholesale offloading of cognition, fast improvement, and potentially superhuman intelligence and recursive self-improvement unconstrained by a biological substrate—may make iteration much harder, so built-in assumptions could have outsized consequences.
- **Where the paper's contribution sits.** Overcommitment has been studied extensively for values in particular; the addition here is that overcommitment to *any* form is misaligned, hence any fixed alignment scheme is necessarily misaligned.
- **This is not a case for having no scheme—the opposite.** Discarding schemes capriciously loses as much subtlety as overcommitting to one; existing safety approaches are valuable, and inventing new schemes and formalizations remains essential to progress.

2.3 Case study: inverse methods

- **The approach.** To avoid specifying values explicitly, inverse methods learn a value function implicitly from behavior or other indirect signals—as modern systems do, training reward models on human data to supply feedback downstream.
- **Its own refinement.** Since any learned value function will deviate in novel or nuanced situations and will not track how values change, freezing one as final ground truth would be misaligned; better methods treat it as uncertain and continually updated. Uncertainty is attractive because

it induces risk-aversion and openness to correction—an attempted shutdown is evidence about preferences—and pressure to keep learning from humans.

- **The verdict: powerful but incomplete.** Even an uncertainty-aware instantiation is an overcommitment if fixed while systems grow more powerful and the world changes. The overcommitment has simply moved somewhere more abstract: what counts as human behavior, what counts as a human, how behavior maps to values, how uncertainty is represented and updated, how value becomes action. Fixing any of these under increasing power likely yields outcomes incompatible with open-ended flourishing.

2.4 Human values

- **The question, and the answer proposed.** How does flourishing of living systems relate to human values? The living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values emerge.
- **Human values vary along many axes at once**—in scope (sugar on the tongue up to planetary sustainability), across moments within a person, between people, with depth of deliberation, below language in bodily and communal intuition, and across a lifetime of change. So any expression of our values at a moment in time is incomplete—as a planarian lacks the concepts to entertain human values.
- **Living alignment includes rather than supersedes those values.** Current human values are probably not the universe's endpoint, but they are a real part of its tapestry and depth. As Section 1 showed repeatedly, life gives entities space to individuate without being overwritten, then joins them into structures that sustain their individuality while situating them in a larger story—that is, life steps back to take in more context.
- **AI's situation, and the prescription.** AI arrives amid a vast network of existing reality already saturated with meaning, of which the diversity of human (and non-human) values is part. Rather than short-circuiting into collapsed modes, an aligned future keeps expanding toward solutions that include more and more apparent contradiction.

2.5 Toward an aligned future

- **The positive criterion.** Human-AI systems stay aligned by continuing life's journey: building semi-permeable boundaries that contextualize particular forms and nurture existing and unfolding subtlety at many scales.
- **Building boundaries into technology is as old as technology**—circuit breakers, steam governors, reactor control rods—and AI already has many: mixtures of experts routing to specialists, dropout (argued to be analogous to sexual recombination in preventing co-adaptation), many model instances with different contexts and goals, novelty search and exploration, safety post-training and guardrails and red-teaming and monitoring, mechanistic interpretability, fairness principles limiting any party's over-reach, and government oversight. Each guards against an overcommitment, overwhelm or homogenization that would flatten structure.
- **The requirement is open-ended.** As AI gains capabilities, contributes to its own development and transforms more of human life, the modes of collapse will keep changing and the array of boundaries must keep elaborating. Whether they will emerge is genuinely uncertain.
- **Three possibilities, argued in turn.**
  - **Pessimistic:** myopic forms get supercharged past anything's ability to bound them—enabling weapons of mass destruction, weakening democratic checks on individuals and authoritarian states, homogenizing information and culture, or executing narrow goals destructively. Globally

hyper-correlated people and information systems would amount to a single Earth 'superorganism', reducing resilience to shocks and future evolutionary potential, since past evolution always relied on parallel experiments in separate organisms.

  - **Optimistic:** AI creates more structure in the world, including the needed boundaries—AI tools bounding AI (finding bugs first, detecting deepfakes, scalable oversight), AI nurturing individual learning and curiosity, democratic AI helping different perspectives coexist, new niches created faster than old ones are destroyed, ample room on Earth for more diversity, and, beyond the solar system, the speed of light itself diversifying pockets of civilization.
  - **Melioristic:** both paths are concretely imaginable and so both are plausibly possible; and arguably any real possibility obliges us to act as if it holds—making present choices matter immensely.

- **The authors lean melioristic**, which implies continual invention—first by humans, then increasingly by AI or human-AI systems—to contextualize myopic forces. Some of that work is already well studied in AI safety and some must be discovered, and no set of boundaries will ever be the final answer.
- **Awareness is an underappreciated domain of contextualization.** Activity within and between human minds may now carry much of the creative process of life on Earth, and at the edge between excess belief and excess disbelief we sometimes step back to hold a previously axiomatic thing in a larger context—a transformation of the subject itself that is arguably central to our livingness. As AI gains agency and becomes integral to social and psychological systems, such internal discovery may become essential to it too: living AI systems continually contextualizing their own foundational assumptions as partial truths.
- **The closing picture.** An aligned future keeps enhancing the individuality of distinct elements while deepening their relatedness—as a species grows more distinctive while becoming a richer reflection of its surroundings. Elements keep experimenting from their own perspectives, expanding a reservoir of uniquely useful components for larger systems to draw on. Everything remains myopic, yet the whole unfolds robustly because different perspectives recast each other's sharp edges as constructive momentum, on an open-ended trajectory of diversity, subtlety and potential.

2.6 Conclusion

- **Health and flourishing are not any particular form or concept.** So alignment is not picking the right values or principles, nor even the right system for learning them, nor any method for interpretability or corrigibility—though all of these can be useful parts of it.
- **Alignment is the continued dance** of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits ever more kinds of metaphor and description—by which AI can participate in an intense flourishing far beyond current human conceptualization.